

LESSON 11.4

INFORMALLY FITTING A LINE TO BIVARIATE DATA



LESSON INTRODUCTION

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

MA.8.DP.1.3 Given a scatter plot with a linear association, informally fit a straight line.

LEARNING TARGETS

- I can describe patterns of associations in data on a scatter plot.
- I can informally fit a straight line to data on a scatter plot.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

MA.912.DP.2.3 Estimate population percentages from data that has been fit to the normal distribution.

ESSENTIAL QUESTIONS

- How can a line of fit model the trend in a scatter plot?
- Where do you place a line of fit in a scatter plot?
- How do you describe patterns of association in a scatter plot?

LESSON NARRATIVE

In the previous lesson, students looked at creating and describing scatter plots. This lesson extends that learning by allowing students to continue describing patterns of associations for scatter plots. Students will also be looking at what makes a good line of fit and how to draw one. They work on interpreting a line of fit and understanding how it can help them interpret the data.

COMPONENT SUMMARY

11.4.1 WARM-UP (~5 MIN.)

Notice and wonder

11.4.3 BRING IT TOGETHER (10-15 MINS.)

Using lines to model bivariate data

11.4.5 YOUR TURN! (5 MINS.)

Practice showing a trend in bivariate data

11.4.2 EXPLORATION (10 MIN.)

Placing a line of fit

11.4.4 GUIDED PRACTICE (10-15 MIN.)

Fitting a line

11.4.6 WRAP-UP (~5 MIN.)

Assess understanding

RESOURCES

GLOSSARY

Key Terms:

- Line of fit
- Outlier

Additional Terms: N/A

MATERIALS

- Rulers, preferably clear
- Uncooked spaghetti
- (Optional) Poster-size graph paper
- (Optional) discs
- (Optional) string

ADDITIONAL PREPARATION

- Have straightedges available for students to use individually.
- If you choose to use any of the optional materials, have them available for students to use in pairs or small groups.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe that there is only one correct placement for a line of fit.
- Students may believe that the stronger the association is, the steeper the slope will be.

THINGS TO CONSIDER

- Have students use different colored pencils to draw multiple, possible lines of fit for one scatter plot. Discuss how each line captures the trend of the data, even though it may not go through every point.
- Remind students that trends are an overall movement and not every point must fit the trend. Point out that outliers are outliers because they fall outside of the general trend.
- If time is an issue, consider assigning 11.4.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Stronger and Clearer Each Time

The lesson invites students to describe the trends and associations in a scatter plot. Demonstrate your thinking out loud about how each graph represents the data and trends. Then, encourage students to reflect on the process, using your example as a bridge to their own thinking.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students will be working to place lines of fit in scatter plots. Clarifications within this standard emphasize expressing connections between concepts and representations. Utilize the progression of the lesson to highlight the process by which students move from a pictorial representation to an understanding of the abstract concept it represents.

DIFFERENTIATE INSTRUCTION

Students are asked quite often in this lesson to explain their thinking or to justify their reasoning. Providing sentence starters, such as “The independent and dependent variables _____” or “The models are trending _____,” may assist ELL students and/or those who need additional assistance.

11.4.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

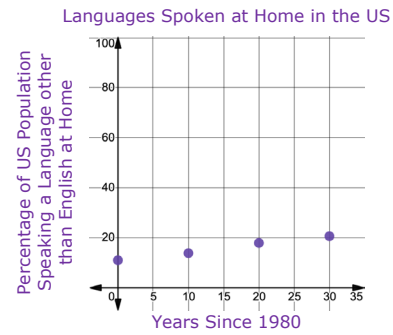
If time allows, there are a lot of interesting questions that could be asked, or that the students might notice/wonder:

- Would it be more appropriate to represent this data as a line graph or scatter plot?
- Why do you think the range of the y-axis is 0-100?
- What do you think these trends looked like before 1980?
- Does speaking a language other than English mean that this percentage of the population doesn't speak English?



11.4.1 Warm-Up: Notice and Wonder

The graph shows the percentage of the US population 5 years of age and older who speak a language other than English at home, according to the U.S. Census, since 1980.



1. What do you notice?

Answers will vary.

Sample response: I notice that the trend of this graph is positive.

2. What do you wonder?

Answers will vary.

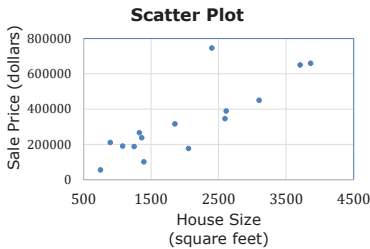
Sample response: I wonder what the most popular language is that people in the US speak other than English?



11.4.2 Exploration: Lines of Fit

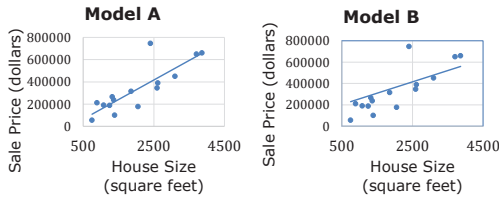
Work with your partner to complete the following.

1. The scatter plot shows the house size and price for the 15 houses sold in Orlando, FL, in 2020.



- a. One of the data points on the scatter plot is (2056, 176900). Interpret this point in context.
This point means that the sale price of a 2,056 sq. ft. house is \$176,900.

Two different lines, or models, are used to represent the relationship between house size and price on the scatter plot. Both models are shown.



- b. What do you notice about the models?
Answers will vary.
Sample response: I notice that both Model A and Model B are trending in a positive direction.
- c. Work with your partner to describe the similarities and differences between the models.

Similarities	Differences
<i>Answers will vary.</i> <i>The independent and dependent variables are the same.</i> <i>Both are trending in a positive direction.</i>	<i>Answers will vary.</i> <i>Model B has more points below the line of fit.</i>

This type of model is called a **line of fit**.



The **line of fit** is a line drawn on a scatter plot to estimate the relationship between two sets of data. It is also known as a trend line.

- d. Discuss with your partner which line appears to better estimate the relationship between house size and sale price. Summarize your discussion.
Answers will vary.
Sample response: the line for Model A appears to be a better estimate because it has more points below and above the line so it would better represent the data.

11.4.2 EXPLORATION: LINES OF FIT

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

Invite students to describe Models A and B along with their similarities and differences. Encourage students to notice the different placements of the lines of fit. Emphasize that different representations can have different purposes and be useful in different situations. Discuss how this concept may apply in the models.

ENRICHMENT

Encourage students to consider the two scatter plots:

- For these scatter plots, why is it called a line of fit?
- How can you determine which line is the best estimate?
- Could we draw a straight line that would hit all of these points?

11.4.3 BRING IT TOGETHER: FITTING LINES TO MODEL BIVARIATE DATA

DIFFERENTIATE INSTRUCTION

Have students recreate the graph using a poster-size piece of graph paper and small discs. Then, give students a piece of string that they can use to represent the line of fit. Have students place the string in multiple positions discussing the pros and cons of each.



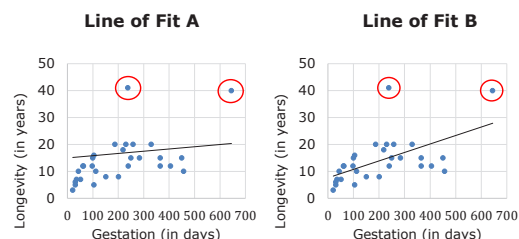
11.4.3 Bring It Together: Fitting Lines to Model Bivariate Data

- Complete the statements.

Most of the data points were below the line in Model ☐ A. ☐ B.

Model ☐ A ☐ B had an approximately equal number of data points above and below the line. Therefore, ☐ A ☐ B better models the relationship between house size and sale price.

- The two scatter plots show the same set of data summarizing the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives. Each scatter plot shows a different line to model the relationship between gestation period and longevity.



TEACHER MOVES

If the pattern of association is positive, the slope is positive. If the pattern of association is negative, then the slope of the line of fit will be negative. It is a common misconception that if there is a strong association, the slope of the line of fit will be steeper. The stronger the association, the more closely the points follow the pattern.

ENRICHMENT

Ensure students understand the significance of outliers by posing the following questions:

- What do the outliers represent?
- Why do you think the outliers occur?
- Is it possible to have a line of fit include the outliers? Why or why not?

- Explain which of the lines is a better fit to the data.

Answers will vary.

Sample response: Line of Fit B is a better fit to the data because it has more points equally spread out above and below the line, where Line of Fit A has more points below the line.

- Complete the statements.

There is a ☐ positive, ☐ negative, ☐ linear ☐ non-linear association

between the gestation period and longevity of an

animal. The association appears to be ☐ weak. ☐ moderate. ☐ strong.

- Discuss with your partner how the slope of the line and fit to the data relates to the patterns of association in a scatter plot that appears linear. Then, summarize your discussion.

Answers will vary.

Sometimes data includes **outliers**.



An **outlier** is a value that is much higher or much lower than the other values in a set of data.

- Circle the point(s) on the scatter plot that appear to be outliers in the data set.

11.4.3 BRING IT TOGETHER: FITTING LINES TO MODEL BIVARIATE DATA (CONTINUED)

e. Complete the statement.

Most of the animals included in the data set follow a similar trend, modeled by the

- ☐ Line of Fit A.
- ☒ Line of Fit B.

However, there

- ☐ is 1
- ☒ are 2
- ☐ are 3

animal(s) included in the

data set who appear to

- ☒ live
- ☐ be pregnant for

much

longer than the others, regardless of

- ☐ how long they live.
- ☒ the length of the gestation period.

11.4.4 GUIDED PRACTICE: INFORMALLY FITTING A LINE

DIFFERENTIATE INSTRUCTION

Use observational data from 11.4.2 Exploration and 11.4.3 Bring It Together to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions shown in the Exploration.

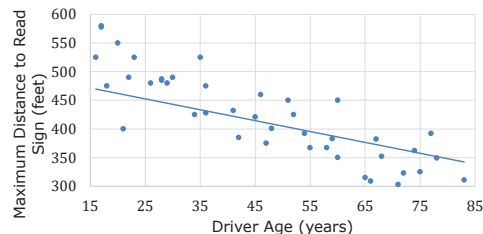
TEACHER MOVES

Consider having students each place their straightedge on their own, and then do a quick gallery walk around the room to see how others have placed theirs (or do this within partners or small groups). Then allow students to revise their placement and use a straightedge to draw their line.



11.4.4 Guided Practice: Informally Fitting a Line

- The scatter plot shows the relationship between driver age and the maximum distance to read a sign.

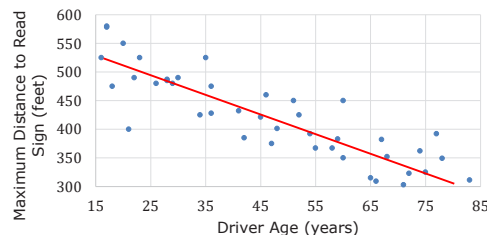


- Explain if the line shown is a good fit for the data.

Answers will vary.

Sample response: the line is a moderate fit for the data because the data is almost equally spread above and below the data, and the data is trending in a negative direction.

- Using a straightedge, draw a line that better fits the data.



- Complete the statement.

As the independent variable ☐ increases, ☐ decreases, the

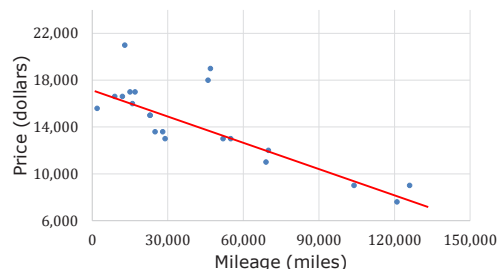
dependent variable tends to ☐ increase. ☐ decrease.

Therefore, there is a ☐ positive ☐ negative pattern of

association between the variables, so the slope of

the line fit to the data is ☐ positive. ☐ negative.

- Wyatt's Auto World sells used cars. The scatter plot displays the data for 20 recently sold used cars by comparing the mileage of the car and its price.



- Using a straightedge, draw a line of fit for the scatter plot.

Answers will vary.

11.4.4 GUIDED PRACTICE: INFORMALLY FITTING A LINE (CONTINUED)

- b. Complete the statement.

The slope of the line of fit is ☐ positive,
☒ negative, meaning

as the mileage on a car ☒ increases,
☐ decreases, the price

Wyatt's Auto World can sell it for ☐ increases.
☒ decreases.

- c. Describe any patterns of association. Use at least two words from the word bank in your description.

decrease(s)	increase(s)	linear
non-linear	positive	negative

Answers will vary.

Sample response: There appears to be a negative linear association between the mileage of a car and its price.

11.4.5 YOUR TURN!

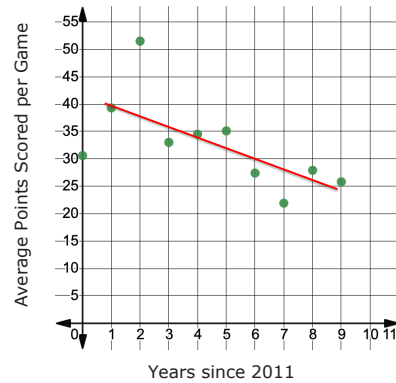
TEACHER MOVES

Have students use a structured pairing strategy as they think about and describe the patterns of association in the graph. Provide students with multiple opportunities to refine and clarify their responses.



11.4.5 Your Turn!

1. The scatter plot shows the average points per game scored by the Florida State University (FSU) Seminoles from 2011-2020.

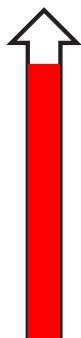


- Use a straightedge to draw a line that models the relationship between the variables.
Answers will vary.
- Describe the patterns of association. Include if any points appear to be outliers in your description.
The slope of the line of fit is negative which means that as the years since 2011 increase, the average number of points scored per game decreases. There also appears to be an outlier 2 years after 2011.

**11.4.6 Wrap-Up: Reflection**

1. Color in the arrow up to the statement that best describes your current understanding.

Answers will vary.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure that I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

11.4.6 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

Use students' reflections to plan future lessons. Consider forming a small group of students who do not thoroughly understand, and gauge which students may be able to progress more quickly through future lessons.